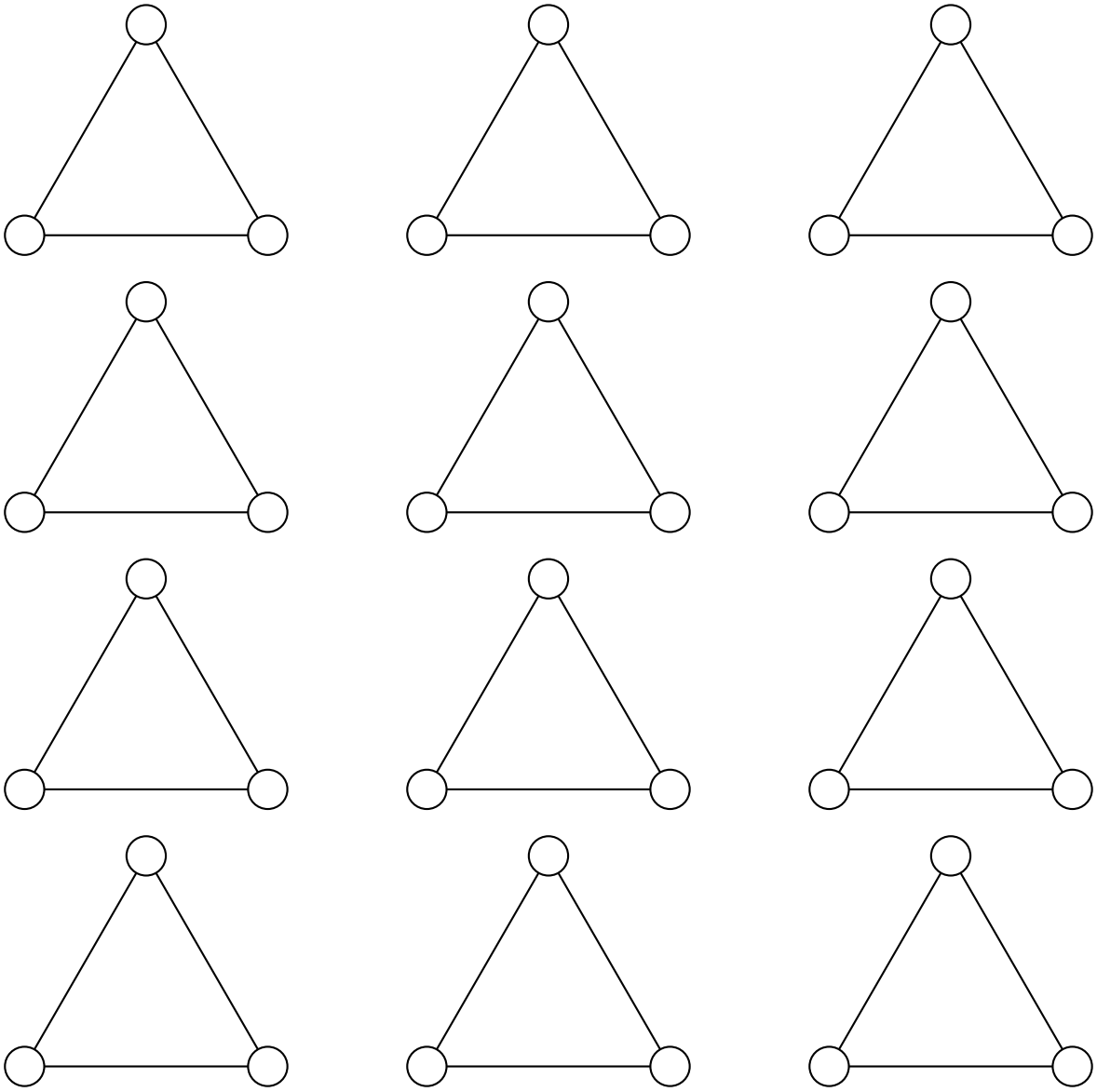
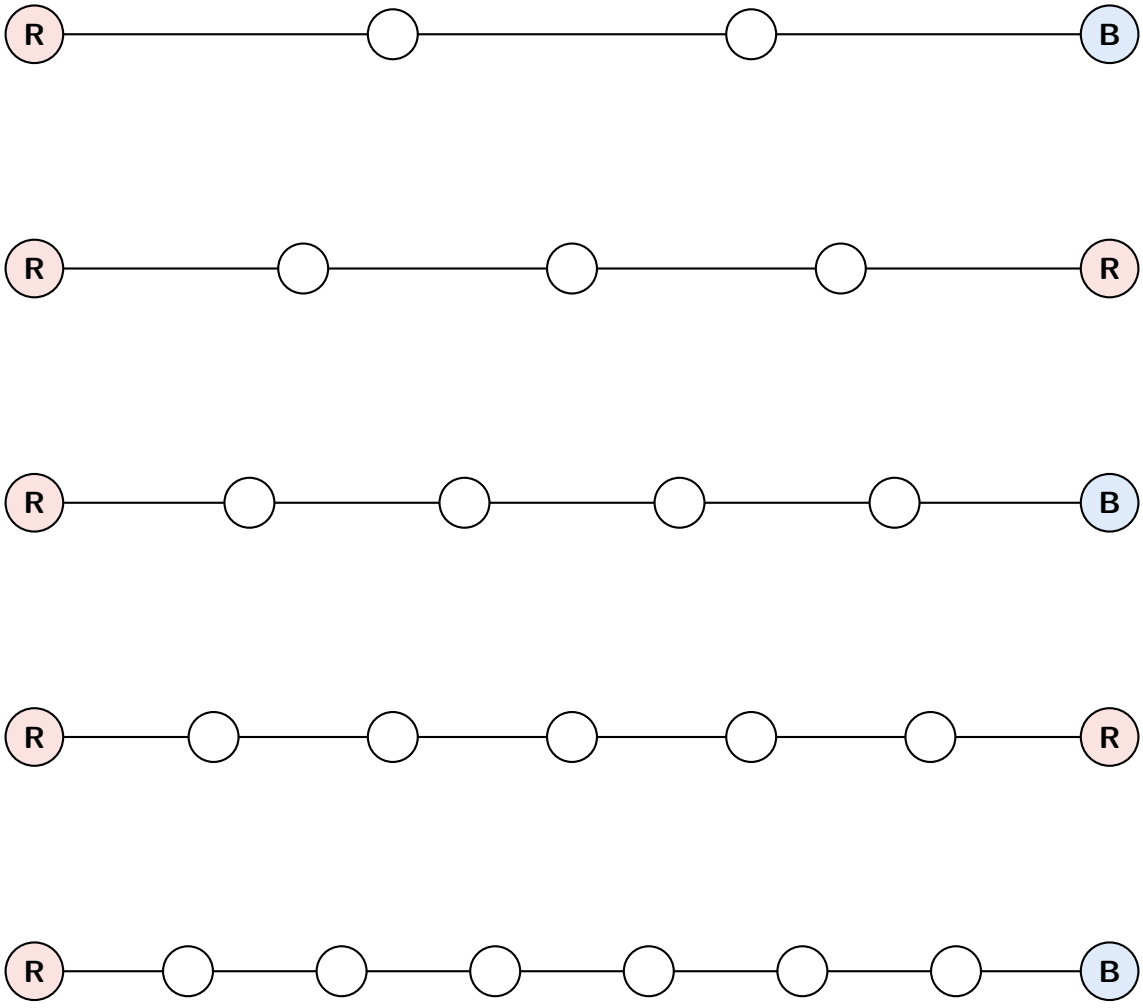
Write R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

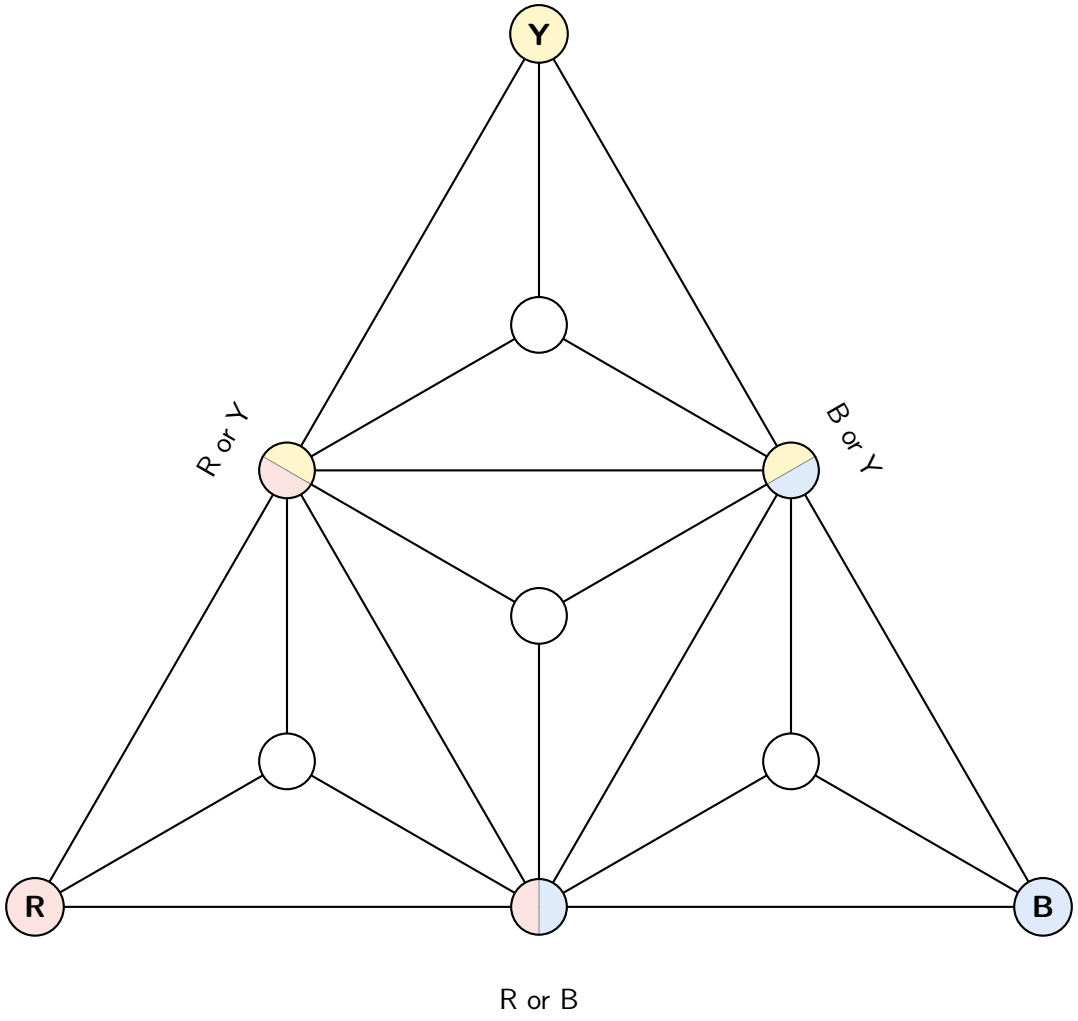
Problem 3: Put R, B, or Y on each dot. Find every different filling, counting turns and flips as the same. Record the number of doors in each filling. Which fillings have exactly one door? Could a triangle have three doors?



Problem 4: Write R or B on every empty dot and mark the doors. Which numbers of doors can each row have? Explain your answer for rows of any length.



Problem 5: Must the number of little triangles with R, B, and Y be odd? Explain your answer for every filling allowed by the side rules, on this board and on any division into little triangles that meet only at corners or along whole edges.



Problem 6: The starred dot may now use R, B, or Y. All other side rules still hold. Find a filling with no little triangle that has R, B, and Y. Explain why this does not contradict your answer to Problem 5.

